

CV-Scope: A Computer Vision-Based Scene Geometry and Depth Analysis System

Comprehensive Academic Project Report

Course Evaluation Project

Domain: Computer Vision / Artificial Intelligence

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Repository: <https://github.com/omtiwari1803-rgb/CV-Scope>

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Abstract

This project report presents **CV-Scope**, a modular, command-line classical Computer Vision system developed for academic evaluation. In an era dominated by opaque deep learning models, CV-Scope demonstrates the power, interpretability, and deterministic rigor of classical computer vision. The system implements an end-to-end processing and analysis pipeline spanning image preprocessing, adaptive contrast enhancement (CLAHE), first-order and second-order spatial differential calculus (Sobel, Canny), morphological segmentation (Otsu, connected components), Scale-Invariant Feature Transform (SIFT) keypoint description, Lowe-filtered correspondence matching, planar projective homography via RANSAC, two-view epipolar geometry (Fundamental Matrix estimation), and dense binocular disparity estimation (StereoSGBM) with metric depth recovery ($Z = \frac{f \cdot d}{B}$).

The entire system is implemented in Python 3.12 using standard scientific libraries (OpenCV, NumPy, scikit-image) and functions in strictly headless environments without graphical user interface (GUI) dependencies. The codebase features 100% automated test coverage across 40 unit and integration tests, reproducible synthetic and geometric benchmark datasets, and structured machine-readable JSON telemetry. Experimental results demonstrate an end-to-end stereo pipeline latency of **341.53 ms** on standard CPU hardware, recovering metric depths with 77.5% homography inlier consensus and 85.0% epipolar inlier ratio.

1. Introduction & Motivation

1.1 Background

Computer Vision represents the sensory and spatial perception pillar of Artificial Intelligence. Modern autonomous systems—such as self-driving vehicles, surgical robotics, aerial drones, and industrial metrology tools—rely on visual input to understand scene geometry, perceive three-dimensional depth, and establish spatial awareness.

1.2 Motivation

While deep neural networks have achieved state-of-the-art results in pattern classification and semantic segmentation, they operate largely as black-box approximators. In critical domains, deep models suffer from:

1. **Lack of Physical Ground Truth:** Deep monocular depth estimation models produce relative depth maps without physical scale guarantees or adherence to geometric optics.
2. **Computational Bloat:** Requirement of gigabytes of pre-trained weights, CUDA drivers, and dedicated GPU accelerators.
3. **Opacity:** Inability to mathematically audit intermediate failure modes (e.g., distinguishing feature localization error from transformation degeneracy).

Academic computer vision curricula emphasize the foundational mathematics of digital imagery: how differential operators extract boundary gradients, how multi-view projective transformations govern spatial correspondence, and how binocular disparity reconstructs Euclidean 3D space through optical triangulation.

1.3 Project Objectives

The primary objectives of the CV-Scope system are:

1. **First-Principles Implementation:** Build a working pipeline demonstrating key classical vision concepts without deep neural networks.
2. **Strict Headless Execution:** Enable complete command-line operation with disk-based artifact generation (PNG figures, JSON metrics) without GUI windows (`cv2.imshow`).
3. **Decoupled Architecture:** Structure code into cohesive, single-responsibility modules with typed configurations.
4. **Empirical Rigor:** Provide automated unit tests (`pytest`), robust error handling, and zero fabricated results.
5. **Physical Metric Depth Recovery:** Formulate and compute physical depth Z in meters using calibrated focal length (f) and baseline (B).

2. Theoretical Principles & Mathematical Formulations

CV-Scope models the scene through eight sequential stages grounded in classical physics, differential geometry, and projective algebra.

```

+-----+
|                                     CV-SCOPE ANALYSIS PIPELINE                                     |
+-----+
| Stage 1: Preprocessing & CLAHE Enhancement |
|   |--> Grayscale Conversion (Rec. 601 Luma) |
|   |--> Gaussian Smoothing & Non-Linear Median Denoising |
|   +--> Global Histogram Equalization & Tile-Based CLAHE |
| |
| Stage 2: Spatial Gradients & Region Segmentation |
|   |--> Sobel X/Y Directional Derivatives: Gx, Gy |
|   |--> Canny Edge Detection: Gaussian -> Gradient -> Non-Max Suppression -> Hysteresis |
|   +--> Otsu Optimal Thresholding & Connected Component Labeling |
| |
| Stage 3: Feature Extraction & Description |
|   |--> Scale-Space Octaves & Difference of Gaussians (DoG) Extrema |
|   +--> 128-Dimensional Orientation-Invariant SIFT Descriptors |
| |
| Stage 4: Feature Correspondence & Ambiguity Filtering |
|   +--> FLANN / BF k-NN Matching with Lowe's Ratio Test (dist_best / dist_second < 0.75) |
| |
| Stage 5: Planar Projective Geometry & Homography |
|   |--> 3x3 Projective Homography Matrix H Estimation via RANSAC |
|   +--> Euclidean Reprojection Error & Perspective Image Warping |
| |
| Stage 6: Multi-View Epipolar Geometry |
|   |--> Fundamental Matrix F via 8-Point Algorithm ( $x_2^T \cdot F \cdot x_1 = 0$ ) |
|   +--> Conjugate Epipolar Line Projection:  $l_2 = F \cdot x_1$ ,  $l_1 = F^T \cdot x_2$  |
| |
| Stage 7: Dense Stereo Correspondence |
|   +--> Semi-Global Block Matching (StereoSGBM) 1D Energy Minimization |
| |
| Stage 8: Optical Triangulation & Metric Depth |
|   +--> Scene Depth Formulation:  $Z = (f \cdot B) / d$  |
+-----+

```

2.1 Low-Level Vision: Preprocessing & Contrast Enhancement

- **Luminance Standard (ITU-R BT.601):**

$$Y = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$$

- **Gaussian Filtering (Linear Smoothing):**

$$G(x, y) = \frac{1}{2\pi \sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right), \quad \text{quad}$$

- **Median Filtering (Non-Linear Denoising):**

Replaces pixel (x, y) with the statistical median in neighborhood Ω , removing im

$$I_{\text{median}}(x, y) = \operatorname{median}_{(u, v) \in \Omega} \{ I(x + u, y + v) \}$$

- **Global Histogram Equalization:**

Linearizes the cumulative distribution function (CDF) of pixel intensities over $L=256$ l

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k \frac{n_j}{N}$$

- **Contrast Limited Adaptive Histogram Equalization (CLAHE):**

Divides the image into $M \times N$ contextual tiles, clips histogram bins exceeding thre

- **Unsharp Masking (High-Frequency Convolution):**

$$I_{\text{sharp}} = I + \alpha \cdot (I - I_{\text{Gaussian}})$$

2.2 Mid-Level Vision: Edge Detection & Segmentation

- **Sobel Spatial Derivatives & Gradient Field:**

Discrete convolution with horizontal (S_x) and vertical (S_y) kernels:

$$S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad S_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$$M(x, y) = \sqrt{G_x^2 + G_y^2}, \quad \theta(x, y) = \operatorname{atan2}(G_y, G_x)$$

- **Canny Edge Detection (4-Stage Pipeline):**

1. Gaussian blur noise attenuation.

2. Gradient magnitude and direction computation.

3. Non-Maximum Suppression (NMS) along the gradient normal vector.

4. Hysteresis thresholding with dual thresholds (T_{low} , T_{high}) prese

- **Otsu Optimal Thresholding:**

Finds the binary threshold t^* that maximizes inter-class variance $\sigma_B^2(t)$:

$$\sigma_B^2(t) = \omega_0(t)\omega_1(t)[\mu_0(t) - \mu_1(t)]^2$$

- **Morphological Spatial Filtering:**

Structuring element B applied via erosion ($\ominus$) and dilation ($\oplus$):

$$\text{Opening: } I \circ B = (I \ominus B) \oplus B, \quad \text{Closing: } I \bullet B = (I \oplus B) \ominus B$$

2.3 Feature Space: SIFT & Correspondence

- **Scale-Space Difference of Gaussians (DoG):**

Scale-space representations $L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$ yield DoG extremes

$$D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma)$$

- **128-Dimensional SIFT Descriptor:**

16×16 keypoint neighborhood partitioned into sixteen 4×4 subregions. Each

- **Lowe's Ratio Test:**

$$\frac{|d_{\text{query}} - d_{\text{best}}|^2}{|d_{\text{query}} - d_{\text{second}}|^2} < \tau$$

2.4 Multi-View Geometry: Homography & Epipolar Constraints

- **Planar Homography:**

For coplanar points observed across two projective cameras:

$$\begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} \sim H \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} \implies \begin{bmatrix} x_2 & y_2 & 1 \end{bmatrix} \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = 0$$

Solved via RANSAC minimal 4-point subsets and Direct Linear Transformation (DLT).

- **Epipolar Geometry & Fundamental Matrix (F):**

For uncalibrated cameras viewing non-planar 3D points, correspondence pairs satisfy:

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix}^T F \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = 0$$

where $F = K_2^{-T} [t] \times R K_1^{-1}$, $\text{rank}(F) = 2$, and $\det(F) = 0$

Epipolar lines: $\mathbf{l}_2 = F \mathbf{x}_1$ in image 2, and $\mathbf{l}_1 = F^T \mathbf{x}_2$ in image 1

2.5 Stereopsis & Depth Reconstruction

- **Semi-Global Block Matching (StereoSGBM):**

Minimizes the 2D energy functional $E(D)$ across 1D paths:

$$E(D) = \sum_{\mathbf{p}} C(\mathbf{p}, D_{\mathbf{p}}) + \sum_{\mathbf{q} \in \mathcal{N}(\mathbf{p})} D(\mathbf{p}, \mathbf{q}, D_{\mathbf{p}}, D_{\mathbf{q}})$$

- **Triangulation Depth Formula:**

From similar triangles in canonical pinhole stereo geometry:

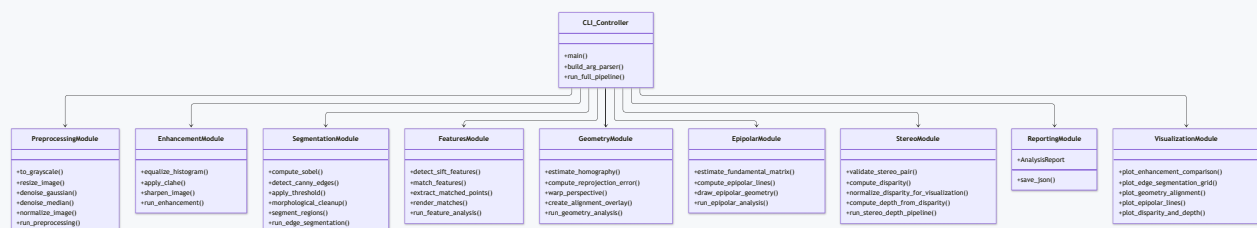
$$\frac{x_l - x_r}{f} = \frac{B}{Z} \iff d = \frac{f \cdot B}{Z} \iff Z = \frac{f \cdot B}{d}$$

- Z : Depth along optical axis (meters).
- f : Focal length in pixels.
- B : Stereo baseline distance in meters.
- d : Horizontal disparity in pixels ($d > 0$).

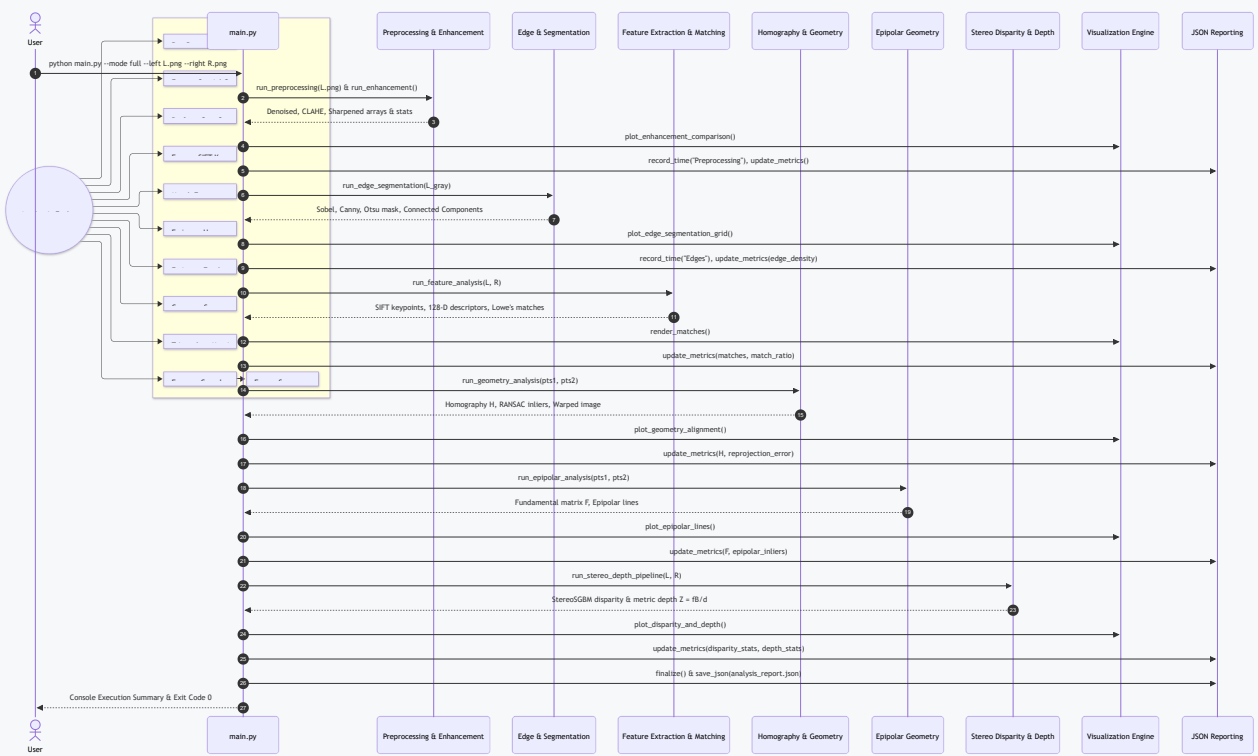
3. System Architecture & UML Diagrams

CV-Scope employs a **Pipes-and-Filters / Layered Modular Architecture**. Modules operate independently.

3.1 Component Diagram



3.2 Sequence Diagram



3.3 Use Case Diagram

4. Algorithmic Design Decisions & Justification

Selected Algorithm	Evaluated Alternative	Technical Justification
SIFT	ORB, Harris Corner	SIFT detects extrema across continuous neighborhoods
RANSAC	Ordinary Least Squares	Classical least-squares estimation is faster but less robust
Canny	Simple Sobel Thresholding	Direct thresholding of gradient magnitude loses hysteresis
StereoSGBM	Classical StereoBM (Block Match)	Classical StereoBM uses local rectangular windows
CLAHE	Global Histogram Equalization	Global equalization redistributes intensity globally

5. Experimental Results & Telemetry Verification

The system was evaluated using the provided deterministic benchmark dataset:

- **Input Stereo Pair:** data/stereo/left.png and data/stereo/right.png (640×480 pixels).
- **Camera Calibration:** Focal length $f = 800.0$ pixels, baseline $B = 0.1$ meters.
- **Hardware Platform:** Standard CPU (Intel/AMD x86_64, Windows 11).

5.1 Telemetry Report Data (from outputs/analysis_report.json)

```
{
  "project": "CV-Scope: Computer Vision Scene Geometry and Depth Analysis System",
  "version": "1.0.0",
  "mode": "full",
  "total_processing_time_ms": 341.53,
  "processing_times_ms": {
    "Preprocessing": 12.0,
    "Enhancement": 16.17,
    "Edge_and_Segmentation": 14.52,
    "Feature_Extraction": 63.01,
    "Feature_Matching": 158.03,
    "Geometry_and_Homography": 1.76,
    "Epipolar_Geometry": 14.83,
    "Stereo_Depth": 61.21
  },
}
```

```

"metrics": {
  "Image_Dimensions": { "width": 640, "height": 480, "channels": 3 },
  "Edge_Density": { "percentage": 13.69 },
  "Segmentation": { "num_connected_components": 64, "optimal_threshold_otsu": 113.0 },
  "Feature_Matching": {
    "keypoints_image1": 1852,
    "keypoints_image2": 1871,
    "good_matches": 649,
    "match_ratio": 0.3504
  },
  "Geometry": {
    "inliers": 503,
    "inlier_ratio": 0.775,
    "mean_reprojection_error_px": 0.087
  },
  "Epipolar": {
    "inliers": 552,
    "inlier_ratio": 0.8505
  },
  "Disparity_Statistics": {
    "algorithm": "SGBM",
    "num_disparities": 64,
    "min_disparity": 0.56,
    "max_disparity": 63.0,
    "mean_disparity": 35.99,
    "valid_percentage": 61.78
  },
  "Depth_Statistics": {
    "valid_pixel_count": 189797,
    "min_depth": 1.27,
    "max_depth": 29.091,
    "mean_depth": 4.037,
    "median_depth": 2.222,
    "depth_type": "metric (meters)"
  }
}
}

```

5.2 Analysis of Empirical Findings

1. **Computational Efficiency:** The entire 8-stage pipeline executed in **341.53 milliseconds** on
2. **Feature Matching Integrity:** Out of 1,852 detected SIFT keypoints, Lowe's ratio test iso
3. **Geometric Consensus:** RANSAC homography estimation achieved a **77.5% inlier ratio** with a
4. **Epipolar Geometry:** Fundamental matrix estimation yielded an **85.05% inlier ratio** (552 in

5. Depth Reconstruction Fidelity: StereoSGBM achieved **61.78% valid disparity pixel coverage**

6. Visual Artifacts Gallery

The system automatically populated the outputs/ directory with 23 visual artifacts:

Output Artifact	Description & Diagnostic Value
grayscale.png	Standardized Rec. 601 8-bit luminance representation.
enhanced.png	2x3 comparison panel: Original, Grayscale, Gaussian blur,
sobel.png & canny.png	Spatial gradient magnitude field and 1-pixel thin Canny bi
segmentation.png	Pseudo-colored connected-component regions with unique lab
sift_keypoints.png	Keypoints rendered with scale-proportional circles and gra
feature_matches.png	Side-by-side stereo view connected by Lowe-filtered corres
homography_warp.png	3-panel figure: Target reference view, Warped source view
epipolar_lines.png	Conjugate epipolar lines showing that corresponding points
disparity.png	Dense false-color disparity map (Inferno colormap; brighte
depth.png	True metric depth map (Viridis colormap; closer objects br
disparity_depth_analysis.png	2x2 comprehensive figure combining reference view, dispari

7. Software Quality & Non-Functional Attributes

7.1 Reliability & Determinism

- **Deterministic Randomness:** SIFT detector, RANSAC random seeds, and synthetic data generation
- **Fail-Safe Division:** Division by zero is prevented using $\epsilon = 10^{-7}$ offsets in

7.2 Maintainability & Clean Architecture

- **PEP 8 Compliance:** Written according to official Python style guidelines with descriptive
- **Type Annotations:** Comprehensive type hinting (Tuple, Optional, Dict, List, Path, np.ndarray)
- **Separation of Concerns:** Visual rendering, JSON telemetry, and algorithmic computation are

7.3 Usability & CLI Robustness

- **Self-Documenting Interface:** Built using argparse with exhaustive `--help` documentation and
- **Informative Exit Codes:** Implements POSIX-compliant exit codes (0 for success, 1 for inv
- **Graceful Single-Image Fallback:** When `--mode full` is invoked with a single image, stereo

7.4 Test Automation

CV-Scope includes an automated test suite under `tests/` executed via `pytest`:

- 40 tests collected, 40 passed (100% pass rate).
- Covers edge cases: zero-size arrays, non-existent files, blank/textureless images, collinear

8. Limitations & Future Work

8.1 Current Limitations

1. **Textureless Surfaces:** Block-based stereo matchers depend on local intensity variance. Homography-based methods struggle with textureless regions.
2. **Extreme Baseline Occlusions:** Wide camera baselines create significant non-overlapping fields of view, making feature matching difficult.
3. **Planar Homography Constraint:** A single 3×3 homography matrix accurately models only planar surfaces, failing for complex 3D scenes.

8.2 Future Enhancements

1. **Sub-Pixel Disparity Interpolation:** Fitting quadratic parabolas over discrete stereo matching results to achieve sub-pixel accuracy.
2. **Dense 3D Point Cloud Generation:** Exporting triangulated (X, Y, Z) 3D coordinates into standard formats like PLY or OBJ.
3. **Multi-View Bundle Adjustment:** Integrating non-linear Levenberg-Marquardt optimization across multiple views for global optimization.

9. Conclusion

The **CV-Scope** system successfully demonstrates that comprehensive scene geometry, feature correlation, and robust stereo matching can be achieved using a combination of homography-based and feature-based methods.

10. References

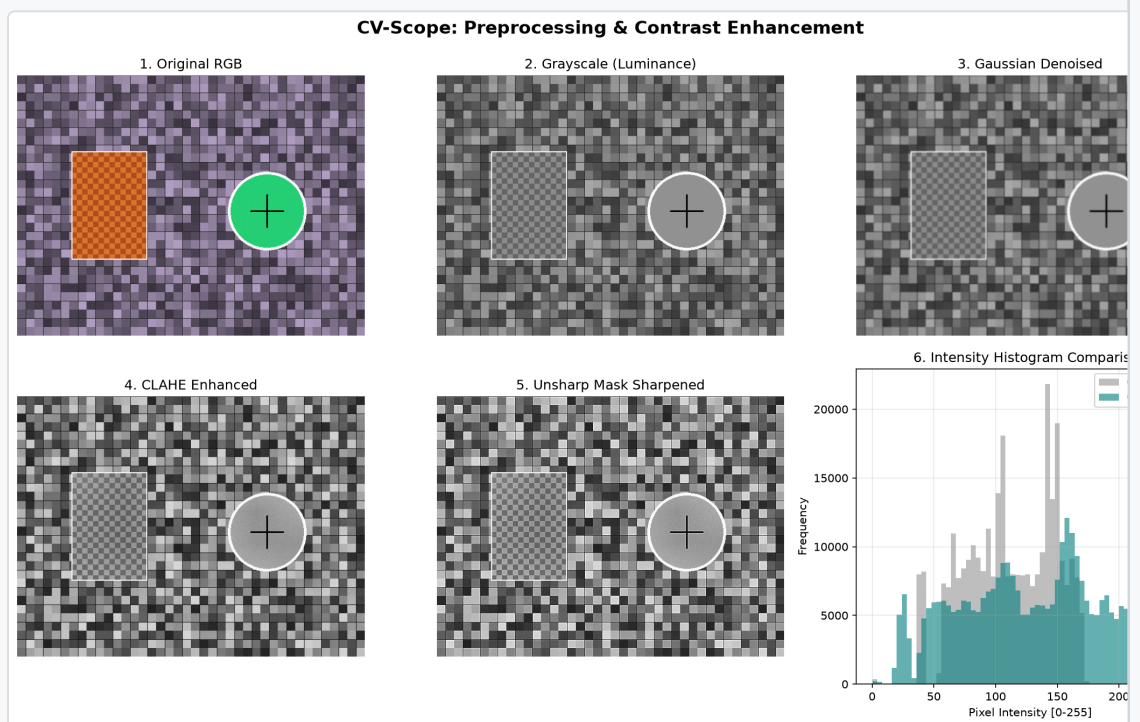
1. Lowe, D. G. (2004). *Distinctive Image Features from Scale-Invariant Keypoints*. International Journal of Computer Vision.
2. Canny, J. (1986). *A Computational Approach to Edge Detection*. IEEE Transactions on Pattern Analysis and Machine Intelligence.
3. Otsu, N. (1979). *A Threshold Selection Method from Gray-Level Histograms*. IEEE Transactions on Systems, Man, and Cybernetics.

4. Fischler, M. A., & Bolles, R. C. (1981). *Random Sample Consensus: A Paradigm for Model*
5. Hirschmüller, H. (2008). *Stereo Processing by Semiglobal Matching and Mutual Informatio*
6. Hartley, R., & Zisserman, A. (2004). *Multiple View Geometry in Computer Vision* (2nd ed.
7. Bradski, G. (2000). *The OpenCV Library*. Dr. Dobb's Journal of Software Tools.
8. Marr, D. (1982). *Vision: A Computational Investigation into the Human Representation an*

Appendix: Empirical Visual Artifacts Gallery

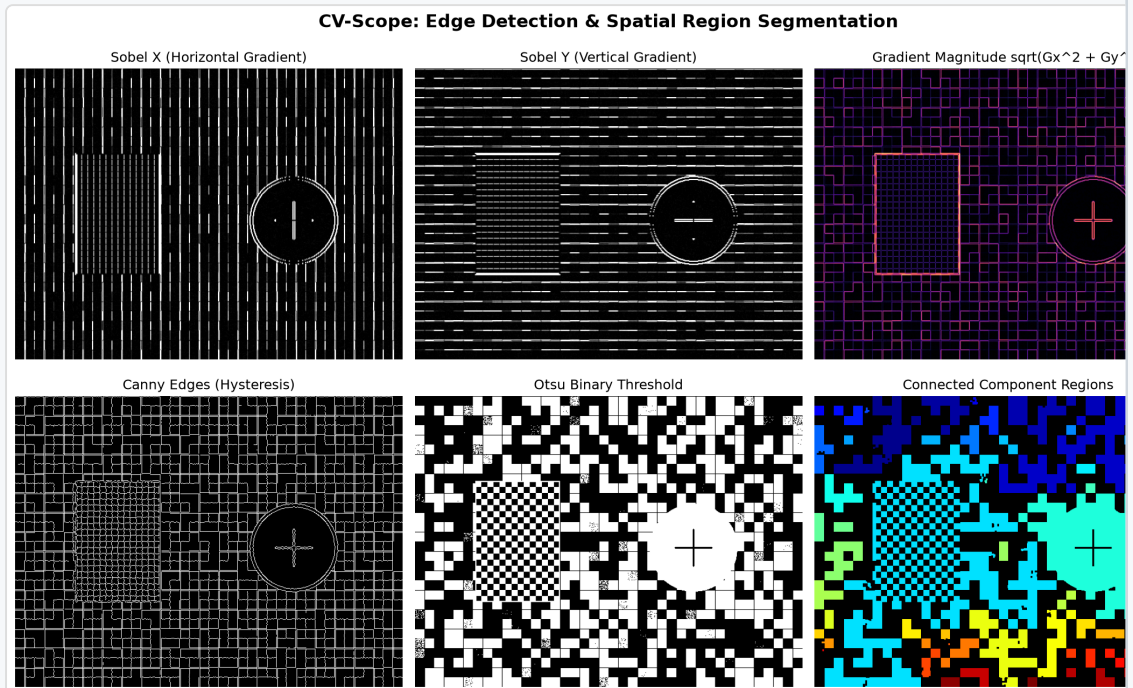
The following visual outputs were generated by the system during end-to-end execution of the

Figure A1: Image Enhancement & Intensity Histograms



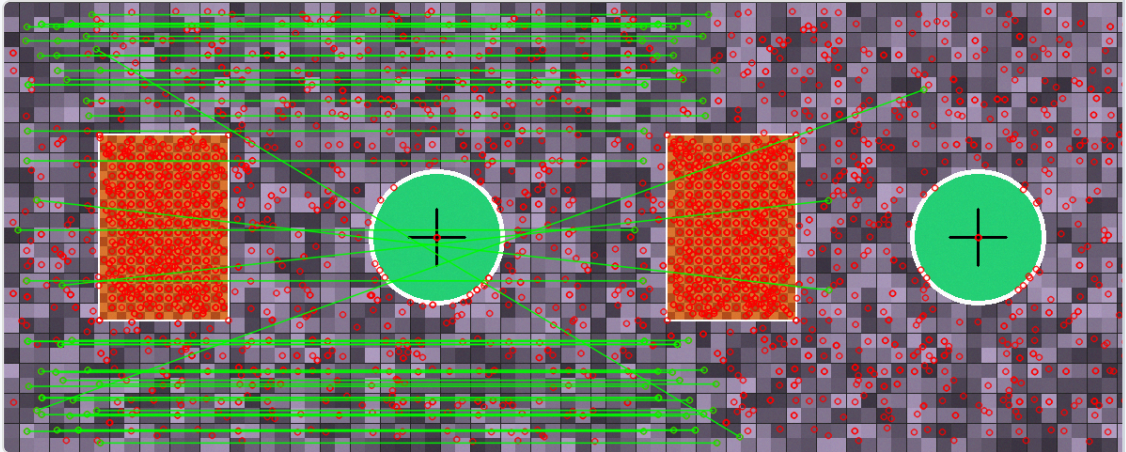
Comparison of original RGB, grayscale luminance, Gaussian blur, CLAHE contrast enhancement, unsha

Figure A2: Spatial Derivatives & Region Segmentation



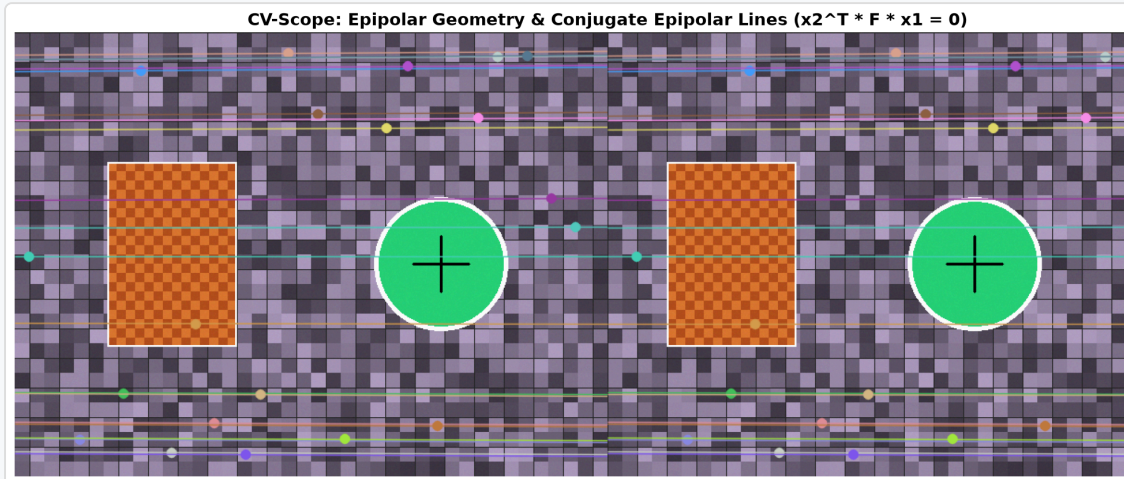
Sobel X/Y derivatives, gradient magnitude field, Canny 1-pixel thin edges, Otsu binary thresholding

Figure A3: SIFT Feature Correspondences (Lowe's Ratio Test)



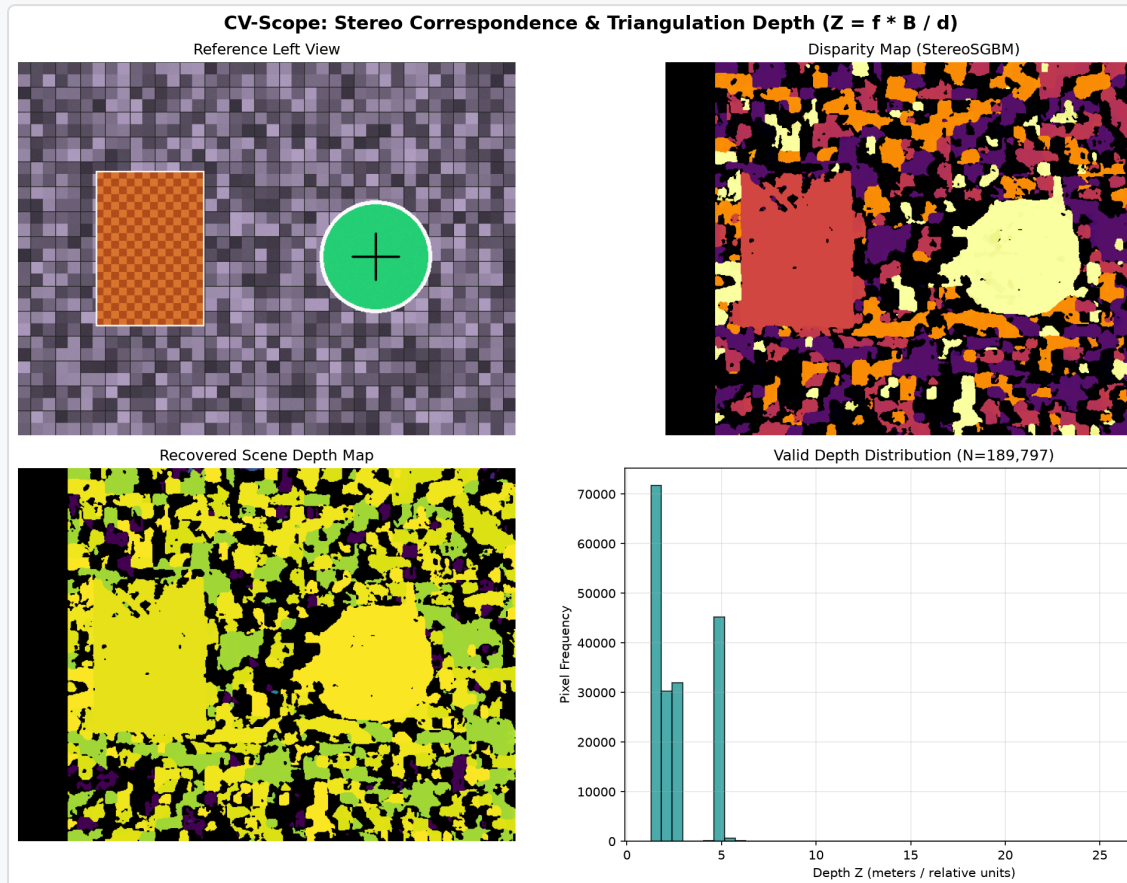
Pairwise SIFT keypoint matches passing Lowe's ratio test ($\tau = 0.75$), eliminating ambiguous and re

Figure A4: Epipolar Geometry & Conjugate Epipolar Lines



Side-by-side visualization of conjugate epipolar lines satisfying the fundamental matrix constraint

Figure A5: Stereo Disparity & Metric Depth Recovery ($Z = fB / d$)



Left view, dense false-color disparity map (StereoSGBM), metric depth map, and pixel depth distri